

HAT-P-27b

R-0080

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_250416 · created 2026-07-15T20:26:38+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460781.8691 +/- 0.0082 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.04
Transit depth (Rp/Rs)^2	1.89 +/- 1.1 [%]
Orbital Inclination (inc)	85.37 +/- 2.45
Airmass coefficient 1 (a1)	1.011 +/- 0.02
Airmass coefficient 2 (a2)	-0.009 +/- 0.062
Scatter in the residuals of the lightcurve fit is	1.98 %
Best Comparison Star	#2 - [468, 459]
Optimal Aperture	3.93
Optimal Annulus	8.63
Transit Duration (day)	0.025 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.04 vs 0.119 (deviation 0.47 σ)
Duration fit vs published	0.025 d vs 0.07 d (deviation 2.8 σ)
Mid-time O-C	-20.0 min
Depth SNR	3.4
Residual scatter	1.98 %

Light curve

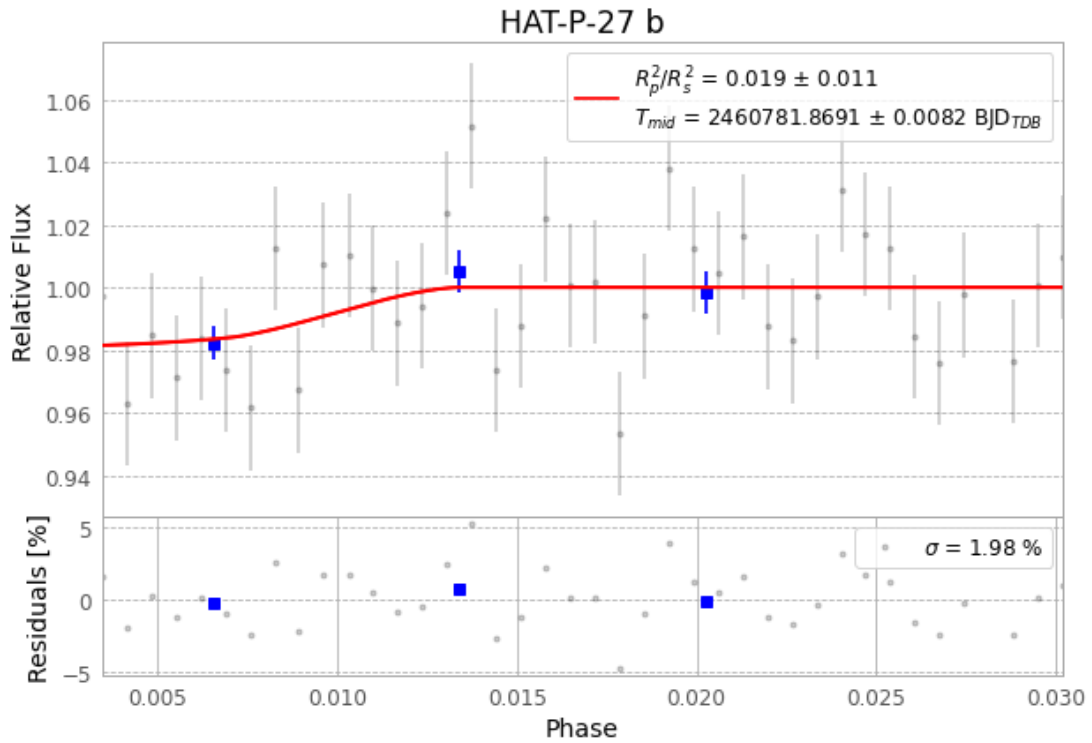


Figure 1 — FinalLightCurve_HAT-P-27 b_2025-04-16.png (SHA-256 fc625f2f6bcf08f6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a09074d040185942...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

40 FITS frames (26 MB), HATP-27250416085715.FITS → HATP-27250416105415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-250416.log (2413ff039d65...), FinalLightCurve_HAT-P-27 b_2025-04-16.png (fc625f2f6bcf...), FinalLightCurve_HAT-P-27 b_2025-04-16.csv (918b9d225a05...)

Compute resources

Wall time 95.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 20:26Z.